Question ID: aad7e1b9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = \frac{1}{10}x - 2$. What is the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane?

Answer

A. $(-2, 0)$

B. $(0, -2)$

C. $\left(0, \frac{1}{10}\right)$

D. $\left(\frac{1}{10}, 0\right)$

Correct Answer: B

Rationale

Choice B is correct. The *y*-intercept of the graph of a function in the *xy*-plane is the point on the graph where $x = 0$. It's given that $f(x) = \frac{1}{10}x - 2$. Substituting 0 for x in this equation yields $f(0) = \frac{1}{10}(0) - 2$, or $f(0) = -2$. Since it's given that $y = f(x)$, it follows that $y = -2$ when $x = 0$. Therefore, the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane is $(0, -2)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.